



ECON42710 Advanced Econometrics: Time Series Assignment

Lecturer: Dr. Vincent Hogan

Student Names: Chou Chia Hui & Cian Donlan

Student ID: 25263607 & 25265334

Module: ECON42710 Advanced Econometrics: Time Series

This report analyses the WTI crude oil price series using both a univariate ARIMA approach and a bivariate VAR approach. The empirical strategy follows the standard time series sequence used in class: visual inspection, formal unit root testing, model identification, estimation, diagnostic checking, forecasting, and forecast evaluation.

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Introduction

This assignment examines the time series properties and forecasting performance of DCOILWTIC0, the WTI crude oil price series, using two econometric approaches.

In Question 1, we estimate a univariate ARIMA model for WTI crude oil prices. The analysis follows the Box–Jenkins approach: graph the series, test for a unit root, identify candidate models from the autocorrelation structure of the differenced series, estimate the competing specifications, assess them using information criteria and residual diagnostics, and then evaluate out-of-sample forecast performance.

In Question 2, we extend the analysis to a vector autoregression (VAR) by adding a second variable, PPIAC0, the Producer Price Index for All Commodities. The VAR framework allows lagged interactions between oil prices and producer prices to be incorporated into the forecasting exercise. We test the stationarity properties of both variables, select the lag length using information criteria, estimate the VAR, examine impulse responses and Granger-causality results, and then compare the VAR forecast with the univariate benchmark.

All analysis was implemented in R, and the full code less some comments is reported in Appendix A. The full R script containing all of the code used will also be submitted alongside this report, including the necessary packages to be installed to conduct the analysis.

Question 1: Univariate ARIMA Forecast

Question 1(a): Plot of the series and visual assessment of stationarity

We first imported the dataset, retained the date variable together with DCOILWTIC0, removed missing values, and plotted the level series against calendar time. The aim of this first step is to assess visually whether the series appears stationary, trend-stationary, or non-stationary with a stochastic trend.

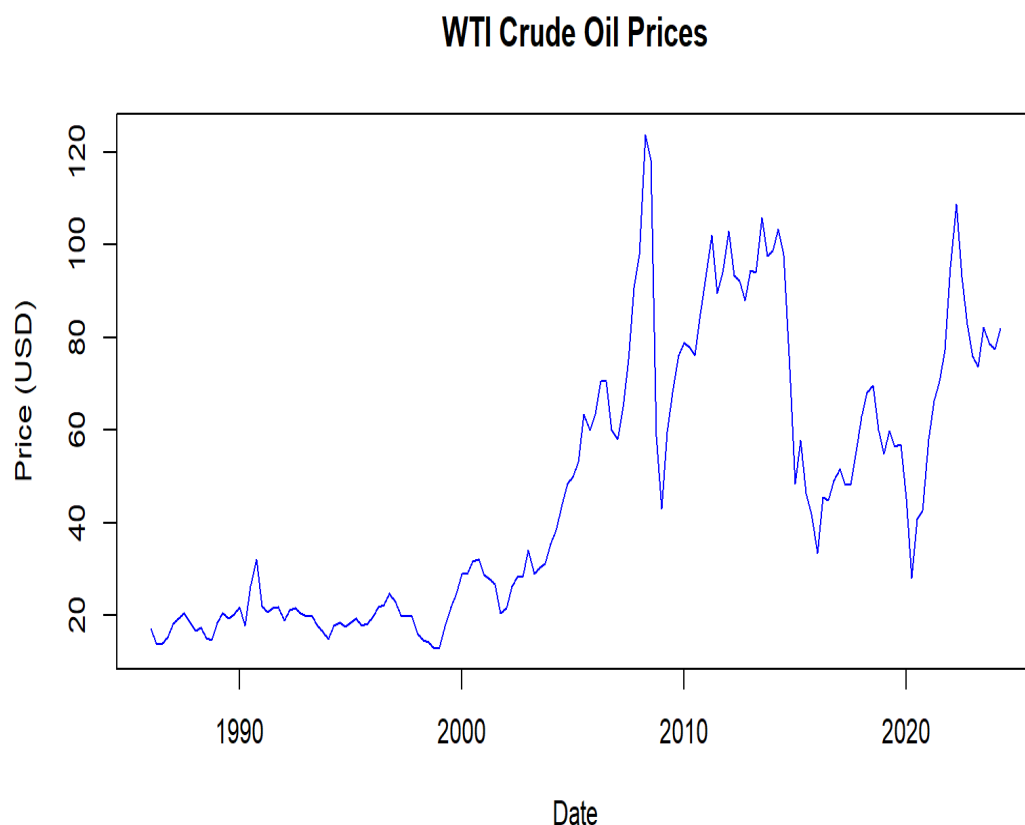


Figure 1: WTI crude oil price series

Figure 1 suggests that the series is non-stationary in levels. Rather than fluctuating around a fixed mean, the series exhibits long, persistent movements and large swings over time. This is more consistent with a stochastic trend or random-walk-type behaviour than with a stationary process or a simple deterministic linear trend.

The corresponding R code is reported in Appendix A, Listing 1.

Question 1(b): Formal unit root test

To test formally for the presence of a unit root, we used the Augmented Dickey–Fuller (ADF) test from the `urca` package. The null hypothesis is that the series has a unit root and is therefore non-stationary in levels; the alternative is that the series is stationary around the deterministic terms included in the test equation.

We estimated two specifications:

- **drift:** includes a constant but no deterministic time trend;
- **trend:** includes both a constant and a linear time trend.

The **drift** specification is the preferred one here. The graph suggests persistent stochastic movements in oil prices, but not a clear deterministic linear trend. A **trend** specification therefore risks over-specifying the testing equation. A **none** specification would also be implausible,

since it would imply a zero long-run mean under the stationary alternative, which is not sensible for an oil price series.

Because the assignment dataset is quarterly, the lag length was selected automatically by AIC from a maximum of four lags, corresponding to one year of quarterly dynamics. The corresponding R code is reported in Appendix A, Listing 2.

Under the preferred `drift` specification, the ADF test statistic is -1.7418 , which is not more negative than the 10% critical value of -2.57 . Hence, we fail to reject the null hypothesis of a unit root. The `trend` specification gives the same conclusion. Therefore, `DCOILWTICO` is non-stationary in levels, and the subsequent ARIMA analysis proceeds using one difference.

Question 1(c): ACF and PACF of the differenced series

Since the level series appears to contain a unit root, we followed the standard Box–Jenkins procedure and examined the autocorrelation function (ACF) and partial autocorrelation function (PACF) of the first-differenced series:

$$\Delta\text{Oil}_t = \text{Oil}_t - \text{Oil}_{t-1}.$$

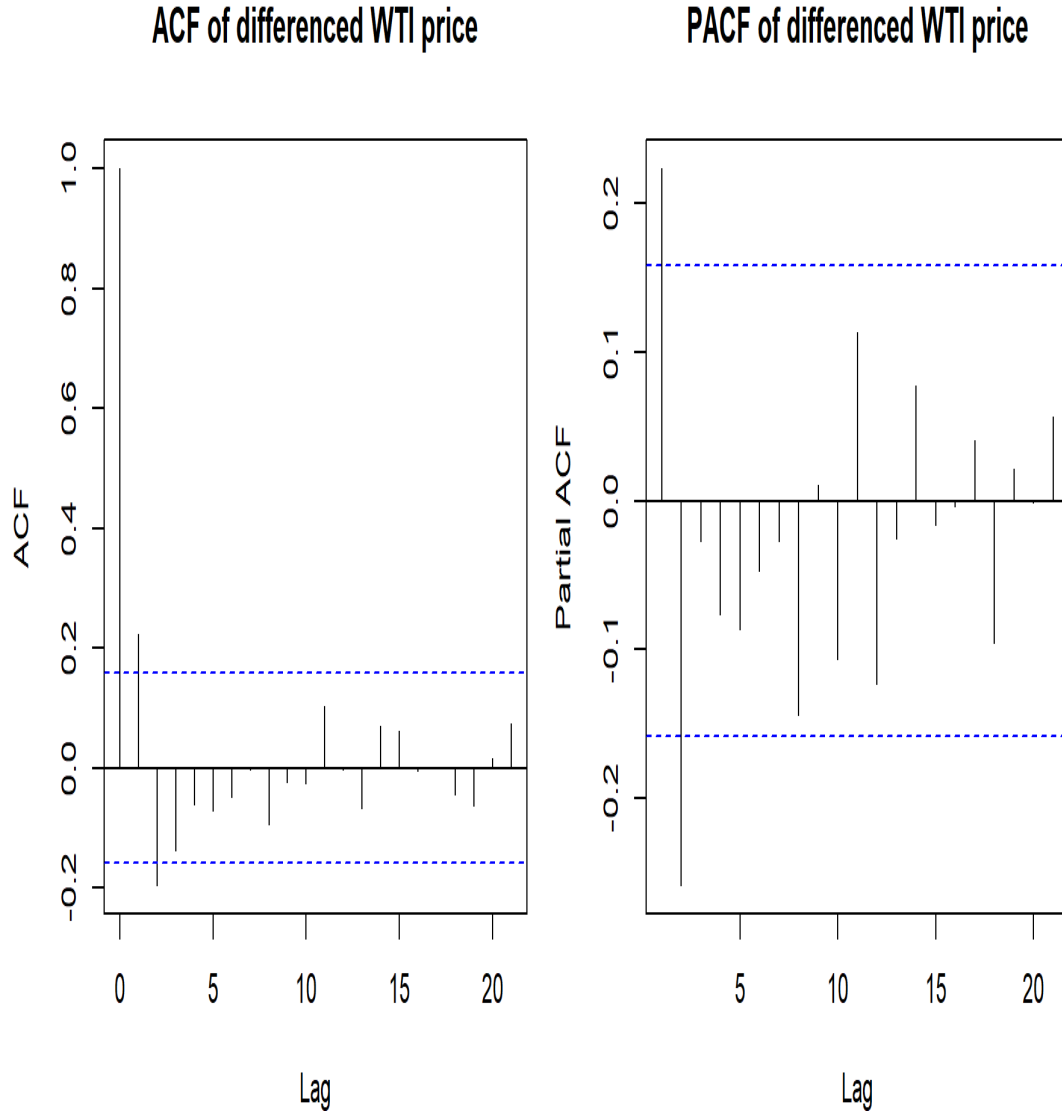


Figure 2: ACF and PACF of the differenced WTI series

The ACF shows significant correlation at the first one or two lags and then cuts off quickly, which is consistent with a low-order MA process in first differences. The PACF also suggests that the short-run dynamics are concentrated at low lags.

On this basis, we selected two plausible candidate models:

$$\text{ARIMA}(0, 1, 2) \quad \text{and} \quad \text{ARIMA}(1, 1, 1).$$

The first corresponds to an MA(2) in differences, while the second allows for an ARMA(1,1) structure in differences.

The corresponding R code is reported in Appendix A, Listing 3.

Question 1(d): Estimation of candidate ARIMA models

We then estimated both candidate models as ARIMA processes with one order of differencing imposed, consistent with the unit root result from part (b):

$$\text{ARIMA}(0, 1, 2), \quad \text{ARIMA}(1, 1, 1).$$

The corresponding R code is reported in Appendix A, Listing 4.

The estimates obtained were:

Table 1: Estimated candidate ARIMA models

Model	Main estimated results
ARIMA(0,1,2)	$ma_1 = 0.2573$, $ma_2 = -0.1705$; $\hat{\sigma}^2 = 65.86$, log-likelihood = -537.53 , AIC = 1081.06
ARIMA(1,1,1)	$ar_1 = -0.2697$, $ma_1 = 0.5755$; $\hat{\sigma}^2 = 66.54$, log-likelihood = -538.30 , AIC = 1082.59

At this stage, both models appear plausible, but ARIMA(0,1,2) is slightly preferred because it has the lower AIC and a marginally lower estimated innovation variance.

Question 1(e): Model assessment and selection

To assess model adequacy, we used three diagnostics:

1. AIC
2. Residual ACF plots
3. Ljung–Box tests for residual serial correlation

The residual ACF plots provide a visual check for remaining serial dependence, while the Ljung–Box test gives a formal test of the null hypothesis that the residuals are white noise.

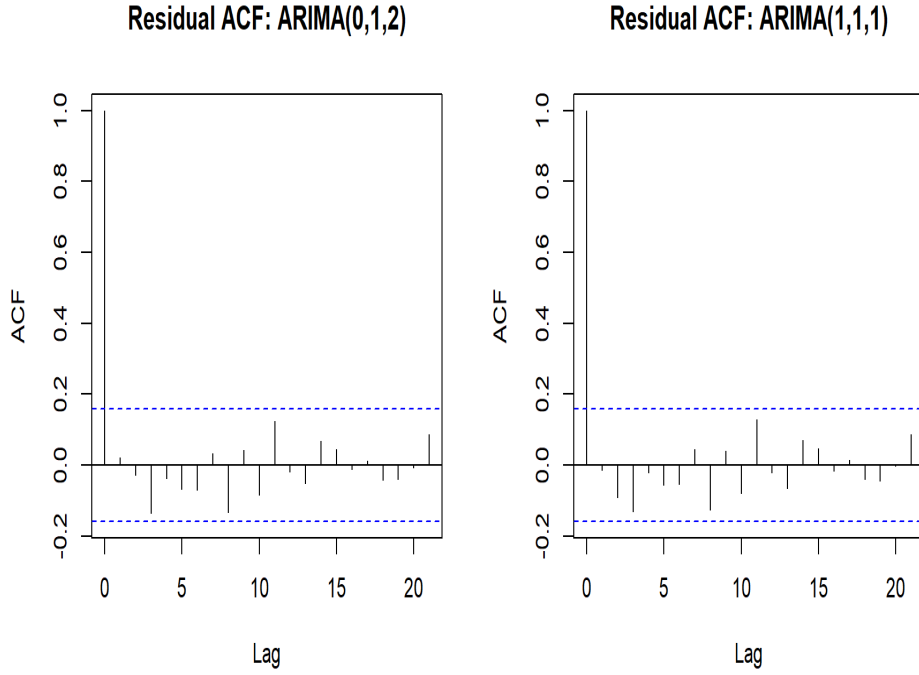


Figure 3: Residual ACF plots for the candidate ARIMA models

The corresponding R code is reported in Appendix A, Listing 5.

The main results are summarised below.

Table 2: Model comparison and residual diagnostics

Model	AIC	Ljung–Box p-value	Conclusion
ARIMA(0,1,2)	1081.06	0.7153	Residuals consistent with white noise
ARIMA(1,1,1)	1082.59	0.6719	Residuals consistent with white noise

Both models pass the residual white-noise test, since in each case the null of no residual serial correlation is not rejected. The residual ACF plots also show no important remaining autocorrelation. This means both models are statistically adequate representations of the differenced dynamics.

However, we select **ARIMA(0,1,2)** as the preferred specification because it has the lower AIC.

Question 1(f): Forecasts for the final year

We used the preferred ARIMA(0,1,2) model to generate out-of-sample forecasts. Specifically, we withheld the final four observations from estimation and treated them as a test sample, corresponding to the final year of quarterly data. The model was estimated on the remaining training sample, and then four-step-ahead forecasts were produced.

Point forecasts and 95% forecast bands were generated using:

$$\hat{y}_{T+h} \pm 1.96 \cdot \text{SE}(\hat{y}_{T+h}).$$

ARIMA(0,1,2): Quarterly Forecast for Final Year

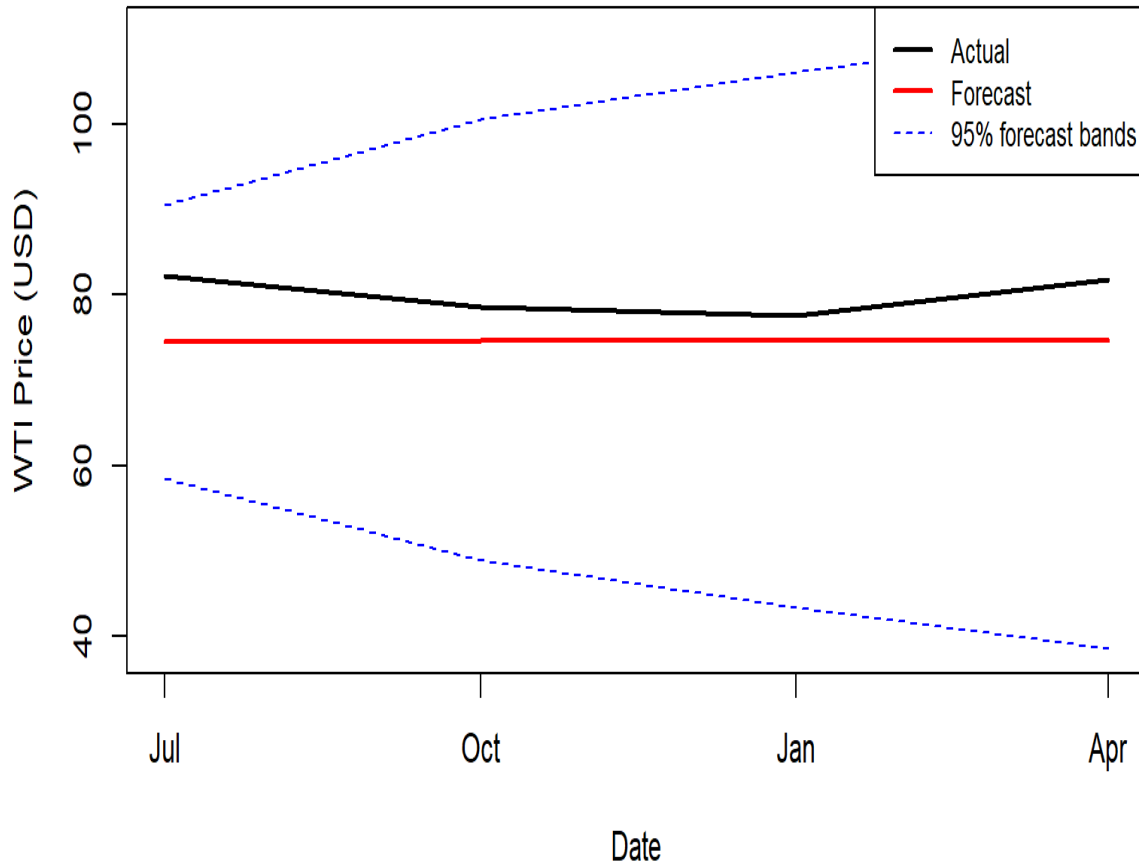


Figure 4: ARIMA(0,1,2) forecast for the final year

The corresponding R code is reported in Appendix A, Listing 6.

The forecast path is fairly smooth, as expected from an ARIMA model with one difference and no drift term. Over a four-quarter horizon the intervals widen with the forecast horizon, but much less dramatically than in the earlier weekly misinterpretation. The model still abstracts from abrupt market moves, so it should be viewed as a conservative benchmark rather than a structural description of oil-price dynamics.

Question 1(g): Forecast evaluation

We evaluated forecast performance using the following standard measures:

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Mean Error (ME), which measures forecast bias

- Empirical coverage of the 95% forecast intervals

The corresponding R code is reported in Appendix A, Listing 7.

The results are:

Table 3: ARIMA(0,1,2) forecast evaluation

Metric	Value
RMSE	5.76
MAE	5.37
Bias (ME)	5.37
95% Band Coverage	100.0%

On the corrected four-quarter holdout, the forecast errors are materially smaller than in the earlier weekly-based calculation. The mean error is positive, indicating that the ARIMA forecast understates oil prices on average over this final year. The 100% empirical coverage result should be interpreted cautiously, since it is based on only four observations and therefore mainly shows that the forecast bands were wide enough to contain the realised path.

Overall, the univariate model provides a reasonably coherent quarterly benchmark, but the short holdout window means its out-of-sample performance should still be interpreted cautiously.

Question 2: VAR Forecast

Question 2(a): Choice of companion variable and economic justification

For the VAR analysis, we selected PPIACO, the Producer Price Index for All Commodities, as the second variable.

This is a plausible companion variable for oil prices for two reasons. First, crude oil is an important production input across many industries, so changes in oil prices can feed through into producer prices. Second, producer prices may also contain information about broader cost pressures and aggregate demand conditions, both of which are relevant for future oil price movements. This makes PPIACO a reasonable variable to include in a VAR for forecasting DCOILWTICO.

We then reloaded the dataset, retained the date variable together with DCOILWTICO and PPIACO, converted the date string to a proper date format, and removed rows with missing observations for either series.

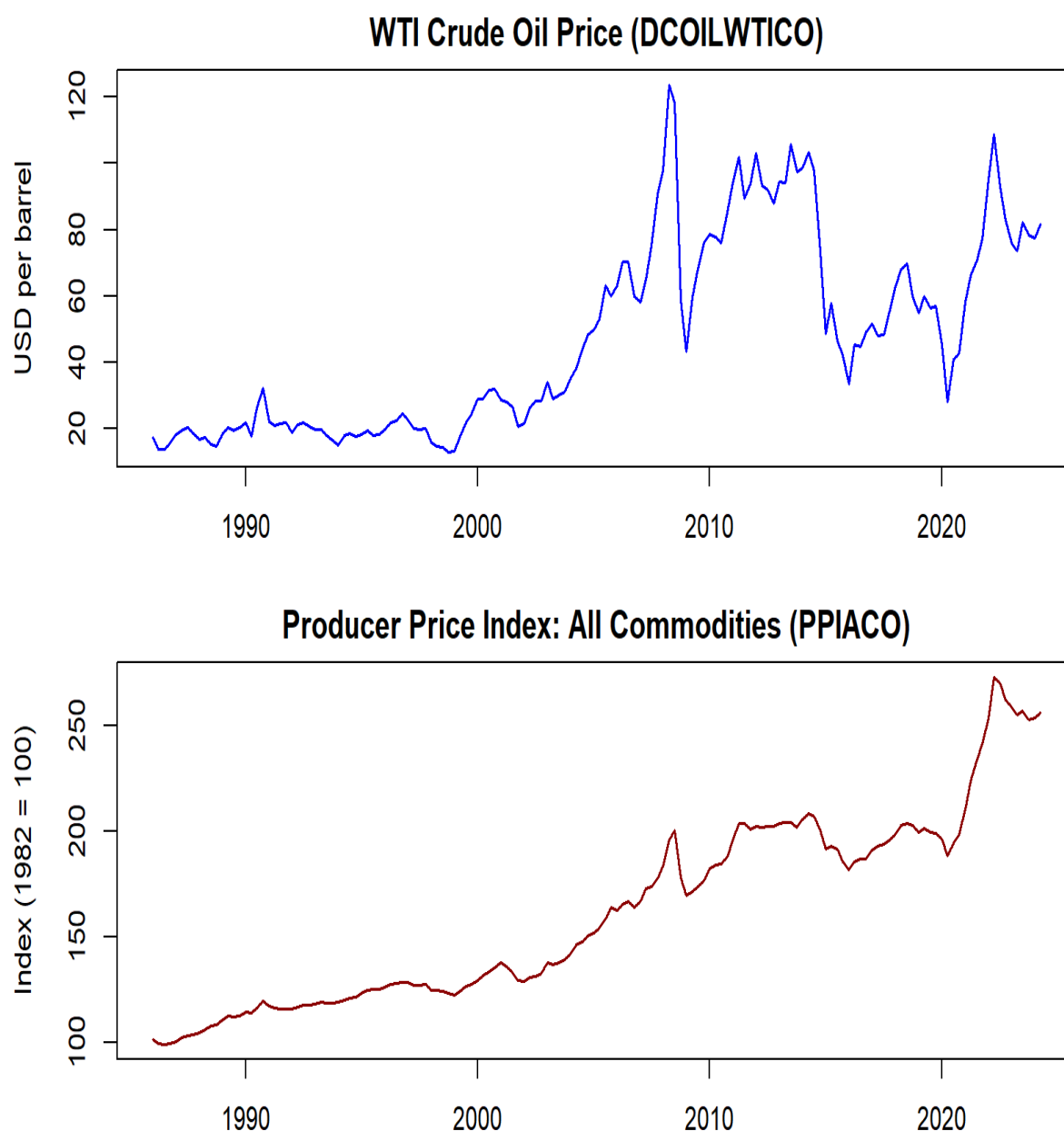


Figure 5: WTI crude oil prices and PPIACO

Figure 5 shows that both series display persistent low-frequency movements rather than stable fluctuations around a fixed mean. This suggests that stationarity should be tested formally before estimating the VAR.

The corresponding R code is reported in Appendix A, Listing 8.

Question 2(b): Stationarity of PPIACO

From Question 1, the oil price series is treated as $I(1)$ for modelling purposes. We therefore next tested PPIACO for a unit root using the Augmented Dickey–Fuller (ADF) test.

We estimated both `drift` and `trend` specifications in levels, and then tested the first-differenced

series. Visual inspection suggests a strong upward movement in `PPIACO`, so the `trend` specification is also relevant here.

The corresponding R code is reported in Appendix A, Listing 9.

Under the `drift` specification in levels, the ADF test statistic is 0.0459, so the null of a unit root is clearly not rejected. Under the `trend` specification in levels, the test statistic is -2.778 , which is not more negative than the 10% critical value of -3.13 , so again the null of a unit root is not rejected. We therefore conclude that `PPIACO` is non-stationary in levels.

We then tested the first-differenced series. For ΔPPIACO , the ADF test statistic under the `drift` specification is -7.7458 , which is far below the relevant critical values, so the null of a unit root is strongly rejected. Hence, `PPIACO` is stationary in first differences and is treated as an $I(1)$ variable.

Since both `DCOILWTICO` and `PPIACO` are $I(1)$, we estimate the VAR in first differences:

$$\Delta\text{Oil}_t, \quad \Delta\text{PPI}_t.$$

A full cointegration analysis is beyond the scope of this assignment, so the VAR is specified directly in first differences.

Question 2(c): Lag-length selection for the VAR

We formed the differenced dataset and then used the `VARselect` function to compare lag lengths from 1 to 8 using the AIC, HQ, and SC (BIC) criteria.

The corresponding R code is reported in Appendix A, Listing 10.

The information criteria do not agree. The AIC selects $p = 8$, while both HQ and SC select $p = 1$. We choose the lag length selected by the AIC, namely $p = 8$. This is a reasonable choice in a forecasting context because the AIC applies a lighter penalty for additional lags, reducing the risk of underfitting the dynamic structure. In time-series forecasting, omitted lag dynamics are often more costly than including a few unnecessary lags. Since the data are quarterly, eight lags also allow the VAR to capture up to two years of interactions between oil prices and producer prices.

Question 2(d): Estimation of the VAR

Using the lag length selected above, we estimated a VAR in first differences with a constant term:

$$Y_t = c + A_1 Y_{t-1} + \cdots + A_p Y_{t-p} + u_t,$$

where

$$Y_t = \begin{pmatrix} \Delta\text{Oil}_t \\ \Delta\text{PPI}_t \end{pmatrix}.$$

The corresponding R code is reported in Appendix A, Listing 11.

The estimated coefficients are reported in Table 4. The results show that both equations contain several statistically significant own-lag and cross-lag terms, indicating non-trivial dynamic interactions between changes in oil prices and changes in producer prices. As is typical in VAR

models, the individual coefficients are less informative than the system as a whole, so the main interpretation will come from the forecasting performance and related dynamic analysis.

Table 4: Estimated VAR(8) coefficients

Lag	Equation for ΔOil_t		Equation for ΔPPI_t	
	ΔOil_{t-k}	ΔPPI_{t-k}	ΔOil_{t-k}	ΔPPI_{t-k}
$k = 1$	0.320**	-0.156	0.157**	0.253
$k = 2$	-0.347**	0.385	-0.138**	0.060
$k = 3$	-0.138	-0.013	0.020	0.054
$k = 4$	-0.163	0.321	-0.141**	0.247
$k = 5$	-0.587***	1.112***	-0.218***	0.420**
$k = 6$	0.136	-1.095***	0.030	-0.504***
$k = 7$	0.208	-0.222	0.077	-0.001
$k = 8$	-0.304**	0.368	-0.067	-0.085
Constant	0.115		0.721*	
R^2	0.269		0.419	
Adjusted R^2	0.178		0.347	
Residual Std. Error	7.981		3.496	

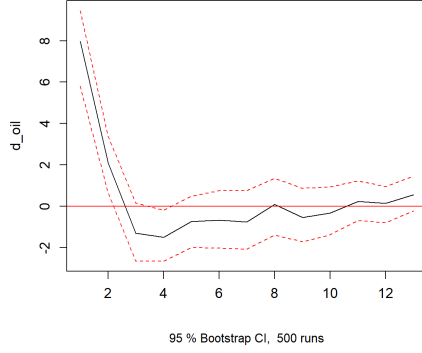
Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Several own-lag and cross-lag coefficients are statistically significant in both equations, suggesting meaningful dynamic interactions between changes in oil prices and changes in producer prices.

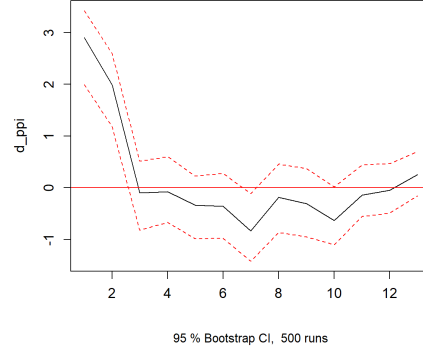
Question 2(e): Impulse response functions

We next computed orthogonalised impulse response functions (IRFs) with 95% bootstrap confidence intervals. The ordering used in the Cholesky decomposition places oil first and PPI second. This means that an oil shock is allowed to affect PPI contemporaneously within the quarter, whereas a PPI shock is not allowed to affect oil contemporaneously. The resulting IRFs therefore depend on this identifying assumption.

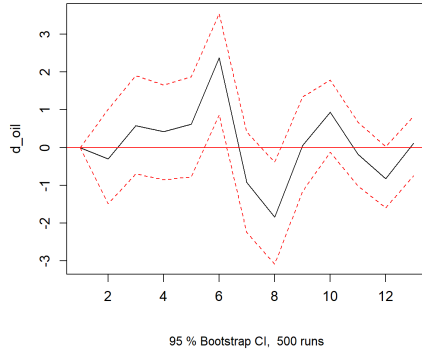
The IRFs trace out the effect over time of a one-off shock to one variable on itself and on the other variable in the system.



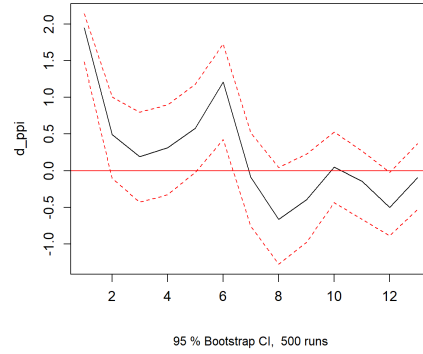
(a) Shock to Oil $\rightarrow$ Oil



(b) Shock to Oil $\rightarrow$ PPI



(c) Shock to PPI $\rightarrow$ Oil



(d) Shock to PPI $\rightarrow$ PPI

Figure 6: Impulse response functions for the estimated VAR

The corresponding R code is reported in Appendix A, Listing 12.

Figure 6 suggests that an oil-price shock has a strong positive own-response on impact, but the response then falls below zero before gradually returning toward zero. This indicates persistence with some oscillation, which is still consistent with a dynamically stable VAR. A positive oil shock also generates a positive initial response in PPI, which is economically plausible through the production-cost channel, although the response becomes smaller and less precisely estimated at longer horizons.

The response of oil prices to a PPI shock appears weaker and more oscillatory, so this channel is harder to interpret confidently. A positive PPI shock does, however, produce a positive own-response on impact, which then declines over time. Overall, the IRFs are broadly economically plausible, but some responses appear imprecisely estimated, especially away from the initial horizons.

We also checked the roots of the estimated VAR. Stability requires all roots to lie inside the unit circle in modulus. This condition is satisfied here, so the impulse responses die out over time rather than explode.

Question 2(f): Granger causality tests

We then performed Granger-causality tests to assess whether lagged values of one differenced variable help to forecast the other.

The null hypothesis in each case is that the lags of the “cause” variable are jointly zero in the relevant equation.

The corresponding R code is reported in Appendix A, Listing 13.

The results reject the null in both directions. For the null that ΔPPI does not Granger-cause ΔOil , the test gives $F = 2.651$ with $p = 0.0083$, so the null is rejected. For the null that ΔOil does not Granger-cause ΔPPI , the test gives $F = 3.5109$ with $p = 0.0007$, so this null is also rejected.

These results indicate bidirectional Granger causality between oil-price changes and producer-price changes. The interpretation is predictive rather than structural: lagged producer-price changes contain useful incremental information for forecasting oil-price changes, while lagged oil-price changes also help forecast producer-price changes.

For the forecasting objective of this assignment, the key implication is that including PPIACO in the VAR is statistically justified, since ΔPPI Granger-causes ΔOil . This indicates that producer-price changes contain incremental predictive information for oil-price changes, although it does not guarantee better out-of-sample forecast performance on a particular holdout sample.

This should not be interpreted as proof of true economic causation. Rather, it is evidence about incremental forecasting content within the VAR.

Question 2(g): VAR forecast of WTI crude oil prices

We next evaluated the out-of-sample forecasting performance of the VAR. We held out the final four observations and estimated the VAR on the preceding sample. The lag length for the forecast VAR was selected on the training sample only. The model then produced four-step-ahead forecasts with 95% forecast intervals.

Because the VAR is estimated in first differences, the oil-price forecasts were initially generated in differences. We then converted the point forecasts back into levels by cumulatively summing predicted oil-price changes from the last available oil-price level in the training sample. The level forecast bands shown are approximate, since they are obtained by cumulatively summing the forecast bounds for oil-price changes.

VAR Forecast vs Actual: WTI Crude Oil Price (Final Year)

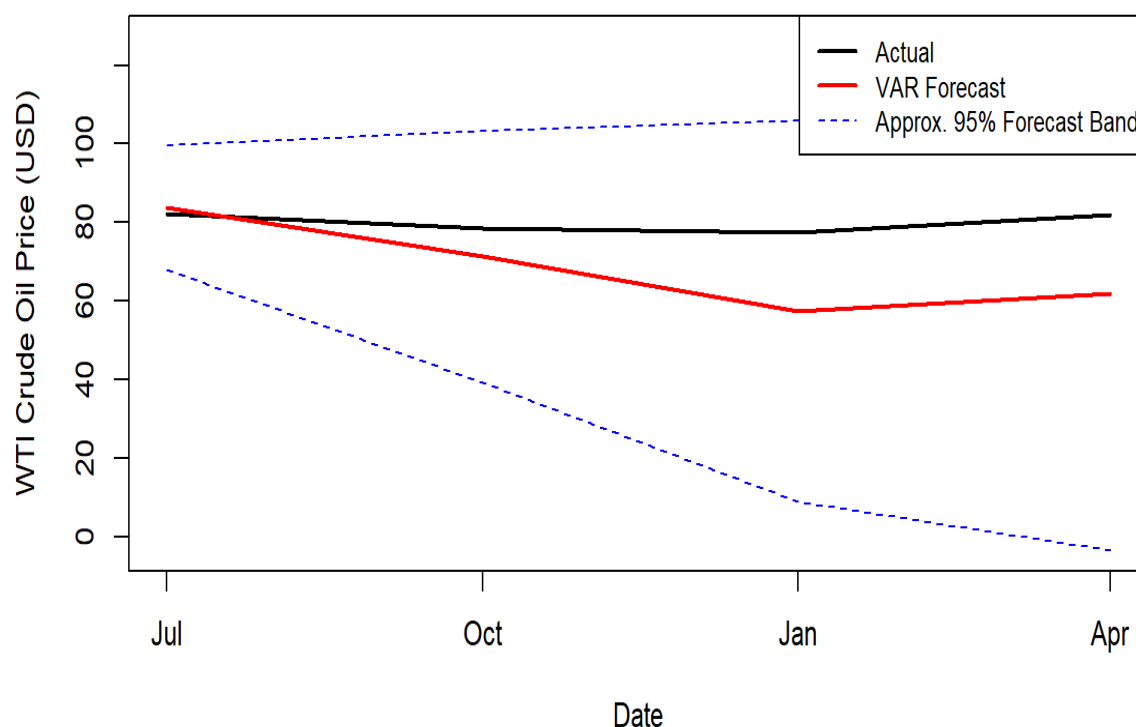


Figure 7: VAR forecast versus actual WTI crude oil prices

The corresponding R code is reported in Appendix A, Listing 14.

Figure 7 compares the actual oil-price path over the holdout period with the VAR forecast and its approximate 95% prediction bands.

Question 2(h): Comparison of VAR and ARIMA forecasts

Finally, we computed the VAR forecast accuracy measures:

- RMSE
- MAE
- Mean Error
- 95% forecast-band coverage

The corresponding R code is reported in Appendix A, Listing 15.

The forecast comparison is reported in Table 5.

Table 5: Forecast comparison: ARIMA versus VAR

Model	RMSE	MAE	Bias (ME)	95% Band Coverage
ARIMA(0,1,2)	5.76	5.37	5.37	100.0%
VAR	14.59	12.14	11.42	100.0%

Using the same final four-quarter holdout for both models, the univariate ARIMA forecast performs better than the VAR on RMSE and MAE. In this corrected comparison, adding PPIACO does not improve point-forecast accuracy for the final year, even though the Granger-causality tests indicate predictive linkages within the full-sample VAR.

However, the interpretation should remain cautious. Both models achieve 100% band coverage over only four holdout observations, so interval calibration cannot be judged precisely. In addition, both forecasts have positive mean error, indicating some tendency to underpredict oil prices over this final year.

Overall, on a like-for-like quarterly comparison, the simpler ARIMA benchmark performs better in point-forecast terms over the holdout sample. The VAR remains econometrically interesting and statistically supported in-sample, but it does not dominate the ARIMA forecast for this particular final-year exercise.

Conclusion

The univariate analysis indicates that WTI crude oil prices are non-stationary in levels and are appropriately treated as an $I(1)$ process. After differencing, the autocorrelation structure suggests a small number of plausible ARIMA specifications. Of the two candidate models considered, ARIMA(0,1,2) is preferred on the basis of AIC and residual diagnostics. However, although it is statistically adequate in-sample, its out-of-sample forecasting performance is limited, particularly during periods of sharp oil-price movement.

The VAR analysis extends the forecasting model by including producer prices. Both series appear to be $I(1)$, so the VAR is estimated in first differences. The model captures dynamic interaction between oil prices and PPI, as reflected in the impulse responses and the bidirectional Granger-causality results. In particular, the Granger-causality tests suggest that producer prices contain useful incremental information for forecasting oil-price changes.

In the corrected forecast comparison based on a common four-quarter holdout, the univariate ARIMA model performs better than the VAR in terms of point-forecast accuracy. This shows that statistically useful multivariate linkages do not automatically translate into better out-of-sample forecasts over a short evaluation window.

Overall, the exercise illustrates the difference between a univariate forecasting approach and a multivariate system approach. The ARIMA model is simpler and uses fewer parameters, while the VAR can capture richer macroeconomic interactions. In this assignment, however, the simpler quarterly ARIMA benchmark performs better over the final-year holdout.

Appendix A: R Code Listings

This appendix contains the R code used in the assignment.

Question 1(a)

Listing 1: R code used for Question 1(a)

```
1 ### Part (a) - Load data, plot, and assess stationarity visually -----
2
3 # Load the dataset
4 import <- read.csv("assignment.csv")
5
6 # Keep dates and the WTI crude oil price column
7 oil <- import[, c("datestr", "DCOILWTICO")]
8
9 # Remove rows with missing values and convert dates
10 oil <- na.omit(oil)
11 oil$datestr <- as.Date(oil$datestr)
12
13 # Plot the raw series to visually assess stationarity
14 plot(oil$datestr, oil$DCOILWTICO, type = "l",
15      main = "WTI Crude Oil Prices",
16      xlab = "Date",
17      ylab = "Price (USD)",
18      col = "blue")
```

Question 1(b)

Listing 2: R code used for Question 1(b)

```
1 ### Part (b) - Formal unit root test (ADF) -----
2
3 # Since the data are quarterly, we allow up to four lags, corresponding to one
4 # year of quarterly dynamics, and let AIC choose the final lag length.
5 summary(ur.df(oil$DCOILWTICO, type = "drift", lags = 4, selectlags = "AIC"))
6 summary(ur.df(oil$DCOILWTICO, type = "trend", lags = 4, selectlags = "AIC"))
```

Question 1(c)

Listing 3: R code used for Question 1(c)

```
1 ### Part (c) - ACF/PACF of differenced series; propose candidate models -----
2
3 # Because the level series contains a unit root, we follow the Box-Jenkins
4 # approach and examine the ACF and PACF of the FIRST-DIFFERENCED series.
5
6 d_oil <- diff(oil$DCOILWTICO)
7
8 par(mfrow = c(1, 2))
9 acf(d_oil, main = "ACF of differenced WTI price")
10 pacf(d_oil, main = "PACF of differenced WTI price")
11 par(mfrow = c(1, 1))
```

Question 1(d)

Listing 4: R code used for Question 1(d)

```
1 ### Part (d) - Estimate candidate ARIMA models -----
2
3 # Both models are estimated in levels with d = 1 imposed (one difference),
4 # consistent with the unit root finding in part (b).
5
6 arima_012 <- arima(oil$DCOILWTICO, order = c(0, 1, 2))
7 arima_111 <- arima(oil$DCOILWTICO, order = c(1, 1, 1))
8
9 print(arima_012)
10 print(arima_111)
```

Question 1(e)

Listing 5: R code used for Question 1(e)

```
1 ### Part (e) - Model selection: information criteria and residual diagnostics --
2
3 # --- Information criteria ---
4 AIC(arima_012, arima_111)
5 BIC(arima_012, arima_111)
6
7 # --- Residual ACF plots (visual check for remaining autocorrelation) ---
8 par(mfrow = c(1, 2))
9 acf(residuals(arima_012), main = "Residual ACF: ARIMA(0,1,2)")
10 acf(residuals(arima_111), main = "Residual ACF: ARIMA(1,1,1)")
11 par(mfrow = c(1, 1))
12
13 # --- Ljung-Box test for residual serial correlation ---
14 # Null hypothesis: residuals are white noise (no remaining autocorrelation)
15 # fitdf = number of estimated ARMA parameters
16 Box.test(residuals(arima_012), lag = 20, type = "Ljung", fitdf = 2)
17 Box.test(residuals(arima_111), lag = 20, type = "Ljung", fitdf = 2)
```

Question 1(f)

Listing 6: R code used for Question 1(f)

```
1 ### Part (f) - Generate forecasts for the final year -----
2
3 # We use the preferred ARIMA(0,1,2) model.
4 # Strategy: estimate on all observations EXCEPT the last 4 quarters (the final
5 # year of quarterly data), then forecast forward 4 steps and compare with
6 # actuals.
7
8 y <- oil$DCOILWTICO
9 dates <- oil$datestr
10 h <- 4 # hold-out horizon: 4 quarterly observations = 1 year
11 n <- length(y)
```

```

12
13 train <- y[1:(n - h)]
14 test  <- y[(n - h + 1):n]
15 test_dates <- dates[(n - h + 1):n]
16
17 # Fit ARIMA(0,1,2) on training sample only
18 fit_012_forecast <- arima(train, order = c(0, 1, 2))
19
20 # Produce 4-step-ahead point forecasts and standard errors
21 fc    <- predict(fit_012_forecast, n.ahead = h)
22 pred  <- as.numeric(fc$pred)
23 lower <- pred - 1.96 * fc$se # 95% lower forecast band
24 upper <- pred + 1.96 * fc$se # 95% upper forecast band
25
26 # Plot actual vs forecast with 95% prediction bands
27 plot(test_dates, test, type = "l", lwd = 2, col = "black",
28       xlab = "Date", ylab = "WTI Price (USD)",
29       main = "ARIMA(0,1,2): Quarterly Forecast for Final Year",
30       ylim = range(c(test, pred, lower, upper)))
31
32 lines(test_dates, pred, col = "red", lwd = 2)
33 lines(test_dates, lower, col = "blue", lty = 2)
34 lines(test_dates, upper, col = "blue", lty = 2)
35
36 legend("topright",
37       legend = c("Actual", "Forecast", "95% forecast bands"),
38       col     = c("black", "red", "blue"),
39       lty     = c(1, 1, 2),
40       lwd     = c(2, 2, 1),
41       cex     = 0.85)

```

Question 1(g)

Listing 7: R code used for Question 1(g)

```

1 ### Part (g) - Assess forecasting power -----
2
3 # Compute out-of-sample forecast errors
4 forecast_errors <- test - pred
5
6 # Standard forecast accuracy metrics
7 rmse_arima <- sqrt(mean(forecast_errors^2)) # Root Mean Squared Forecast Error
8 mae_arima  <- mean(abs(forecast_errors))   # Mean Absolute Error
9 me_arima   <- mean(forecast_errors)        # Mean Error (bias)
10
11 # Empirical coverage of the 95% forecast bands
12 coverage_arima <- mean(test >= lower & test <= upper)
13
14 # Print results
15 cat("--- ARIMA(0,1,2) Forecast Evaluation ---\n")
16 cat("RMSE      :", round(rmse_arima, 4), "\n")
17 cat("MAE       :", round(mae_arima, 4), "\n")
18 cat("Bias (ME):", round(me_arima, 4), "\n")

```

```
19 cat("95% Band Coverage:", round(coverage_arima * 100, 1), "%\n")
```

Question 2(a)

Listing 8: R code used for Question 2(a)

```
1 # Load the full dataset again, keeping both DCOILWTICO and PPIACO
2 import2 <- read.csv("assignment.csv")
3 df <- import2[, c("datestr", "DCOILWTICO", "PPIACO")]
4 df$datestr <- as.Date(df$datestr)
5
6 # Keep only rows where BOTH series have observations
7 df <- na.omit(df)
8
9 cat("Q2 Sample period:", as.character(min(df$datestr)),
10     "to", as.character(max(df$datestr)), "\n")
11 cat("Number of observations:", nrow(df), "\n")
12
13 # Plot both series for a visual overview
14 par(mfrow = c(2, 1), mar = c(3, 4, 2, 1))
15
16 plot(df$datestr, df$DCOILWTICO, type = "l", col = "blue", lwd = 1.5,
17      main = "WTI Crude Oil Price (DCOILWTICO)",
18      xlab = "", ylab = "USD per barrel")
19
20 plot(df$datestr, df$PPIACO, type = "l", col = "darkred", lwd = 1.5,
21      main = "Producer Price Index: All Commodities (PPIACO)",
22      xlab = "Date", ylab = "Index (1982 = 100)")
23
24 par(mfrow = c(1, 1))
```

Question 2(b)

Listing 9: R code used for Question 2(b)

```
1 cat("\n--- ADF test for PPIACO in levels (drift) ---\n")
2 summary(ur.df(df$PPIACO, type = "drift", lags = 4, selectlags = "AIC"))
3
4 cat("\n--- ADF test for PPIACO in levels (trend) ---\n")
5 summary(ur.df(df$PPIACO, type = "trend", lags = 4, selectlags = "AIC"))
6
7 # If we fail to reject H0 (unit root) in levels, test on first differences:
8 d_ppi_test <- diff(df$PPIACO)
9
10 cat("\n--- ADF test for delta(PPIACO) (drift) ---\n")
11 summary(ur.df(d_ppi_test, type = "drift", lags = 4, selectlags = "AIC"))
```

Question 2(c)

Listing 10: R code used for Question 2(c)

```
1 # We work with first differences of both series since both appear to be I(1).
2 d_oil_var <- diff(df$DCOILWTICO)
3 d_ppi_var <- diff(df$PPIACO)
4
5 # Combine into a matrix (required format for the vars package)
6 var_data <- cbind(d_oil_var, d_ppi_var)
7 colnames(var_data) <- c("d_oil", "d_ppi")
8
9 # VARselect computes AIC, BIC (SC), and HQ for lag lengths 1 to 8
10 lag_select <- VARselect(var_data, lag.max = 8, type = "const")
11
12 cat("\n--- VAR Lag Selection Criteria ---\n")
13 print(lag_select$criteria)
14 cat("\nAIC selects:", lag_select$selection["AIC(n)", "lags\n")
15 cat("BIC selects:", lag_select$selection["SC(n)", "lags\n")
16 cat("HQ selects:", lag_select$selection["HQ(n)", "lags\n")
17
18 # We use the lag length chosen by AIC for the following reasons:
19 #   - AIC applies a lighter penalty for additional lags than BIC/SC, which
20 #     reduces the risk of under-fitting dynamic structure.
21 #   - For forecasting applications, AIC-selected models tend to outperform
22 #     more parsimonious BIC-selected models.
23
24 p_chosen <- lag_select$selection["AIC(n)"]
25 cat("Chosen lag order p =", p_chosen, "\n")
```

Question 2(d)

Listing 11: R code used for Question 2(d)

```
1 # Estimate VAR(p) in first differences with a constant included
2 var_model <- VAR(var_data, p = p_chosen, type = "const")
3
4 cat("\n--- VAR Model Summary ---\n")
5 summary(var_model)
6
7 # Extract the individual equation objects for formatted output
8 eq_oil <- var_model$varresult$d_oil
9 eq_ppi <- var_model$varresult$d_ppi
10
11 # Produce a formatted coefficient table using stargazer
12 # Change type = "text" to type = "latex" when compiling the PDF report
13 cat("\n--- Formatted coefficient table ---\n")
14 stargazer(eq_oil, eq_ppi,
15           title = "VAR Estimates: WTI Oil Price and PPI (first differences)",
16           dep.var.labels = c("Delta Oil Price", "Delta PPI"),
17           type = "text")
```

Question 2(e)

Listing 12: R code used for Question 2(e)

```

1  ### Part (e) - Impulse Response Functions (IRFs) -----
2
3  # Compute orthogonalised IRFs with 95% bootstrap confidence intervals.
4  # n.ahead = 12 quarters (3 years); runs = 500 bootstrap replications.
5  # Ordering: d_oil first, d_ppi second (Cholesky decomposition).
6
7  irf_oil_to_oil <- irf(var_model, impulse = "d_oil", response = "d_oil",
8                        n.ahead = 12, boot = TRUE, runs = 500, ci = 0.95)
9
10 irf_oil_to_ppi <- irf(var_model, impulse = "d_oil", response = "d_ppi",
11                       n.ahead = 12, boot = TRUE, runs = 500, ci = 0.95)
12
13 irf_ppi_to_oil <- irf(var_model, impulse = "d_ppi", response = "d_oil",
14                       n.ahead = 12, boot = TRUE, runs = 500, ci = 0.95)
15
16 irf_ppi_to_ppi <- irf(var_model, impulse = "d_ppi", response = "d_ppi",
17                       n.ahead = 12, boot = TRUE, runs = 500, ci = 0.95)
18
19 # Save each IRF separately at higher resolution for LaTeX
20 png("2_e_1.png", width = 1600, height = 1200, res = 200)
21 par(mar = c(4, 4, 1.2, 1), cex.axis = 1.1, cex.lab = 1.1)
22 plot(irf_oil_to_oil, main = "")
23 dev.off()
24
25 png("2_e_2.png", width = 1600, height = 1200, res = 200)
26 par(mar = c(4, 4, 1.2, 1), cex.axis = 1.1, cex.lab = 1.1)
27 plot(irf_oil_to_ppi, main = "")
28 dev.off()
29
30 png("2_e_3.png", width = 1600, height = 1200, res = 200)
31 par(mar = c(4, 4, 1.2, 1), cex.axis = 1.1, cex.lab = 1.1)
32 plot(irf_ppi_to_oil, main = "")
33 dev.off()
34
35 png("2_e_4.png", width = 1600, height = 1200, res = 200)
36 par(mar = c(4, 4, 1.2, 1), cex.axis = 1.1, cex.lab = 1.1)
37 plot(irf_ppi_to_ppi, main = "")
38 dev.off()
39
40
41 cat("\n--- VAR Stability Check ---\n")
42 print(roots(var_model))
43 # All roots should be < 1 in modulus for a stable VAR.

```

Question 2(f)

Listing 13: R code used for Question 2(f)

```

1  cat("\n--- Granger Causality: Does d_ppi Granger-cause d_oil? ---\n")
2  granger_ppi_to_oil <- causality(var_model, cause = "d_ppi")
3  print(granger_ppi_to_oil$Granger)
4

```

```

5 cat("\n--- Granger Causality: Does d_oil Granger-cause d_ppi? ---\n")
6 granger_oil_to_ppi <- causality(var_model, cause = "d_oil")
7 print(granger_oil_to_ppi$Granger)

```

Question 2(g)

Listing 14: R code used for Question 2(g)

```

1 ### Part (g) - VAR forecast of WTI crude oil price -----
2
3 # Hold out last 4 quarters
4 h_var <- 4
5
6 # Split differenced VAR data into training and test sets
7 n_diff <- nrow(var_data)
8 train_var <- var_data[1:(n_diff - h_var), , drop = FALSE]
9
10 # Choose lag length on TRAINING sample only
11 lag_select_train <- VARselect(train_var, lag.max = 8, type = "const")
12 p_train <- lag_select_train$selection["AIC(n)"]
13
14 # Re-estimate VAR on training sample
15 var_train <- VAR(train_var, p = p_train, type = "const")
16
17 # Forecast h_var steps ahead with 95% intervals
18 var_fc <- predict(var_train, n.ahead = h_var, ci = 0.95)
19
20 # Forecasts for oil-price CHANGES
21 fc_d_oil <- var_fc$fcst$d_oil[, "fcst"]
22 fc_lower <- var_fc$fcst$d_oil[, "lower"]
23 fc_upper <- var_fc$fcst$d_oil[, "upper"]
24
25 # Convert point forecasts back to LEVELS
26 last_level_train <- df$DCOILWTICO[nrow(df) - h_var]
27 pred_levels <- last_level_train + cumsum(fc_d_oil)
28
29 # Approximate level bands
30 lower_levels <- last_level_train + cumsum(fc_lower)
31 upper_levels <- last_level_train + cumsum(fc_upper)
32
33 # Actual holdout-period levels and dates
34 actual_levels <- df$DCOILWTICO[(nrow(df) - h_var + 1):nrow(df)]
35 actual_dates <- df$datestr[(nrow(df) - h_var + 1):nrow(df)]
36
37 # Plot forecast vs actual
38 plot(actual_dates, actual_levels, type = "l", lwd = 2, col = "black",
39       ylim = range(c(actual_levels, pred_levels, lower_levels, upper_levels), na.rm =
40                     TRUE),
41       xlab = "Date", ylab = "WTI Crude Oil Price (USD)",
42       main = "VAR Forecast vs Actual: WTI Crude Oil Price (Final Year)")
43 lines(actual_dates, pred_levels, col = "red", lwd = 2)
44 lines(actual_dates, lower_levels, col = "blue", lwd = 1, lty = 2)

```

```

45 lines(actual_dates, upper_levels, col = "blue", lwd = 1, lty = 2)
46
47 legend("topright",
48       legend = c("Actual", "VAR Forecast", "Approx. 95% Forecast Band"),
49       col     = c("black", "red", "blue"),
50       lty     = c(1, 1, 2),
51       lwd     = c(2, 2, 1),
52       cex     = 0.85)

```

Question 2(h)

Listing 15: R code used for Question 2(h)

```

1  ### Part (h) - Compare VAR forecast vs ARIMA(0,1,2) from Question 1 -----
2
3  # Compute VAR forecast accuracy metrics
4  var_errors <- actual_levels - pred_levels
5  rmse_var   <- sqrt(mean(var_errors^2))
6  mae_var    <- mean(abs(var_errors))
7  me_var     <- mean(var_errors)
8  coverage_var <- mean(actual_levels >= lower_levels & actual_levels <= upper_levels)
9
10 cat("\n--- VAR Forecast Evaluation ---\n")
11 cat("RMSE      :", round(rmse_var, 4), "\n")
12 cat("MAE       :", round(mae_var, 4), "\n")
13 cat("Bias (ME):", round(me_var, 4), "\n")
14 cat("95% Band Coverage:", round(coverage_var * 100, 1), "%\n")
15
16 # ARIMA(0,1,2) results from Question 1 for comparison
17 # These are now computed on the same final four-quarter holdout used for the
18 # VAR forecast, so the comparison is like-for-like.
19 cat("\n--- Side-by-Side Comparison ---\n")
20 cat(sprintf("%-20s %10s %10s %12s\n", "Model", "RMSE", "MAE", "Bias"))
21 cat(sprintf("%-20s %10.4f %10.4f %12.4f\n", "ARIMA(0,1,2)", rmse_arma, mae_arma, me_
    arma))
22 cat(sprintf("%-20s %10.4f %10.4f %12.4f\n", "VAR",          rmse_var,  mae_var,  me_
    var))

```

Appendix B: AI Use Declaration

AI assistance was used to help improve the presentation of the R code, organise the report structure, and refine the wording of the econometric explanations. All modelling choices were made by the authors, and all code was executed, outputs checked, and results interpreted by the authors.